

Equicontinuous Convergence Need Not Have the Lebesgue Property

Automatic mathematics run `fa_banach_001`

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Status. Candidate counterexample, likely valid, for human review. It gives a negative answer to the general continuous-operator question on printed page 8 of Zabetti's paper. The domain is locally convex and has the *AM* and Levi properties; the range is order complete and Lebesgue. The failure is caused only by the equicontinuous convergence topology.

1 Source question

The source is O. Zabetti, *AM-spaces from a locally solid vector lattice point of view with applications*, arXiv:1912.00141, later published in the *Bulletin of the Iranian Mathematical Society* [1]. Theorem 5 of that paper proves that if X has the *AM* and Levi properties, Y is order complete, and Y has the Lebesgue property, then the lattice of order-bounded operators $B^b(X, Y)$ has the Lebesgue property for the topology of uniform convergence on bounded sets. On printed page 8 the author asks whether a version holds for continuous order-bounded operators with the equicontinuous convergence topology. The paper proves the positive result when X is Fréchet (Corollary 7).

In general, we do not know if a version Theorem 5 holds for continuous operators with respect to the equicontinuous convergence topology. But we can have the following partial observation. Suppose X is a Fréchet space which possesses *AM* and Levi properties, then by Proposition 4 and [1 Theorem 5.19], $B^b(X, Y) = B_c^b(X, Y) = B_b^b(X, Y)$ for any order complete locally solid vector lattice Y . So, by using Theorem 5 we have the following.

Corollary 7. *Suppose X is a Fréchet space which possesses *AM* and Levi properties and Y is an order complete locally solid vector lattice. If Y has the Lebesgue property, then so is $B_c^b(X, Y)$.*

Thus the natural general question is:

If X and Y are locally solid vector lattices, X has the *AM* and Levi properties, Y is order complete and has the Lebesgue property, must $B_c^b(X, Y)$ have the Lebesgue property for equicontinuous convergence?

Here a net (S_α) converges equicontinuously to zero if, for every zero neighborhood V in Y , there is a zero neighborhood U in X such that, for every $\varepsilon > 0$, eventually

$$S_\alpha(U) \subseteq \varepsilon V.$$

2 Idea of the proof

Use an uncountable coordinate set I . Give the finite-support lattice $\mathbb{R}^{(I)}$ every weighted ℓ^1 seminorm. This makes bounded sets extremely small: each bounded set uses only finitely many coordinates. The domain therefore has both the *AM* and Levi properties.

For each finite $F \subset I$, delete the coordinates in F and map the remaining finite-support vector into $\ell^1(I)$. These positive continuous operators decrease to zero, because each individual input has finite support. Equicontinuous convergence would require one domain neighborhood on which all sufficiently late coordinate deletions are uniformly small. A neighborhood uses only finitely many weight functions. On an uncountable set, their combined reciprocal cannot tend to zero outside finite sets. Hence every finite deletion leaves a coordinate vector of one fixed positive size inside the neighborhood, and its image has the same fixed ℓ^1 norm.

3 The counterexample

Let I be an uncountable set and put

$$X = \mathbb{R}^{(I)} = \{x \in \mathbb{R}^I : \text{supp}(x) \text{ is finite}\},$$

with coordinatewise order. For every weight $a = (a_i)_{i \in I} \in [0, \infty)^I$ define

$$p_a(x) = \sum_{i \in I} a_i |x_i|.$$

The sum is finite for every $x \in X$. Let τ be the locally convex topology generated by all seminorms p_a . Since each p_a is a lattice seminorm and the coordinate weights separate points, (X, τ) is a Hausdorff locally convex locally solid vector lattice.

Lemma 1. *A subset $B \subset X$ is τ -bounded if and only if there is a finite $J \subset I$ such that $B \subset \mathbb{R}^J$ and B is bounded in that finite-dimensional space.*

Proof. The reverse implication is immediate. Conversely, suppose that the union of the supports of elements of B is infinite. Choose $x_n \in B$ and pairwise distinct $i_n \in I$ with $x_n(i_n) \neq 0$. Define one weight by

$$a_{i_n} = \frac{n}{|x_n(i_n)|}, \quad a_i = 0 \quad (i \notin \{i_n : n \geq 1\}).$$

Then $p_a(x_n) \geq n$, contradicting boundedness. Thus all supports lie in a finite J . The coordinate seminorms show that every coordinate of B is bounded, which is equivalent to finite-dimensional boundedness. $\square$

Lemma 2. *The locally solid vector lattice X has the *AM* and *Levi* properties.*

Proof. Let B be bounded. By Lemma 1, all its elements have support in one finite set J and are coordinatewise bounded. Every finite lattice supremum of elements of B has the same two properties, so $B^\vee$ is bounded. This is the *AM* property.

If $D \subset X_+$ is bounded and upward directed, it is contained in $\mathbb{R}^J$ for some finite J . Its coordinatewise supremum is finite in each coordinate and belongs to $\mathbb{R}^J \subset X$. It is the supremum of D , which proves the *Levi* property. $\square$

Let $Y = \ell^1(I)$ with its norm topology and coordinatewise order.

Lemma 3. *The Banach lattice Y is order complete and has the Lebesgue property.*

Proof. If a nonempty subset of Y is bounded above by $u \in Y$, its coordinatewise supremum differs from any fixed member by a positive vector dominated by an ℓ^1 vector. Hence the supremum belongs to Y . Thus Y is order complete.

Now let $y_\beta \downarrow 0$ in Y . Fix β_0 . The support of y_{β_0} is countable, and $0 \leq y_\beta \leq y_{\beta_0}$ for $\beta \geq \beta_0$. Given $\eta > 0$, choose a finite $K \subset I$ for which

$$\sum_{i \notin K} y_{\beta_0}(i) < \frac{\eta}{2}.$$

Coordinatewise decrease to zero and directedness give $\beta_1 \geq \beta_0$ such that

$$\sum_{i \in K} y_\beta(i) < \frac{\eta}{2} \quad (\beta \geq \beta_1).$$

Therefore $\|y_\beta\|_1 < \eta$ eventually. This is the Lebesgue property. $\square$

Direct the finite subsets $F \subset I$ by inclusion and define

$$T_F : X \longrightarrow Y, \quad (T_F x)_i = \begin{cases} 0, & i \in F, \\ x_i, & i \notin F. \end{cases}$$

Theorem 4. *Every T_F is positive, continuous, and order bounded, and*

$$T_F \downarrow 0 \quad \text{in } B_c^b(X, Y).$$

However, (T_F) does not converge to zero in the equicontinuous convergence topology. Consequently $B_c^b(X, Y)$ does not have the Lebesgue property.

Proof. Positivity is clear, and every positive linear operator is order bounded. If $\mathbf{1}_{I \setminus F}$ is the 0–1 weight of the complement of F , then

$$\|T_F x\|_1 = p_{\mathbf{1}_{I \setminus F}}(x),$$

so T_F is continuous. If $F \subset G$, then $0 \leq T_G \leq T_F$. If a positive operator S is a lower bound for every T_F , then, for each $i \in I$, choosing $F = \{i\}$ gives

$$0 \leq S e_i \leq T_{\{i\}} e_i = 0.$$

Since the coordinate vectors span X , $S = 0$. Hence $T_F \downarrow 0$ in the operator lattice.

It remains to disprove equicontinuous convergence. Let

$$V = \{y \in Y : \|y\|_1 < 1\}.$$

Take an arbitrary zero neighborhood U in X . It contains a basic neighborhood

$$W = \{x : p_{a^j}(x) < \delta_j \text{ for } j = 1, \dots, m\},$$

where the a^j are nonnegative weights and $\delta_j > 0$. For every $i \in I$ put

$$b_i = \max \left\{ 1, \frac{a_i^1}{\delta_1}, \dots, \frac{a_i^m}{\delta_m} \right\}.$$

Every b_i is finite and

$$I = \bigcup_{M=1}^{\infty} \{i \in I : b_i \leq M\}.$$

Because I is uncountable, for some integer M the set $A_M = \{i : b_i \leq M\}$ is infinite. For every finite $F \subset I$, choose $i \in A_M \setminus F$ and set $x = (2M)^{-1}e_i$. Then, for each j ,

$$p_{aj}(x) = \frac{a_i^j}{2M} \leq \frac{\delta_j}{2},$$

so $x \in W \subset U$. But $i \notin F$, and therefore

$$\|T_F x\|_1 = \frac{1}{2M}.$$

With $\varepsilon = (4M)^{-1}$, this shows $T_F(U) \not\subset \varepsilon V$ for every finite F . Thus there is no zero neighborhood U satisfying the defining eventual-containment condition for equicontinuous convergence. $\square$

Remark 5. *The uncountable direct-sum topology is not metrizable, so the counterexample does not conflict with the source paper’s positive Frechet result. The proof also identifies the precise obstruction: one fixed neighborhood cannot encode decay along uncountably many coordinates modulo finite sets.*

4 Verification and limitations

The construction is purely exact and uses no numerical computation. The hypotheses were audited separately: Lemma 1 proves the needed bounded-set structure, Lemma 2 verifies both domain properties, Lemma 3 treats arbitrary decreasing nets in the range, and Theorem 4 checks both order convergence and the literal quantifiers in equicontinuous convergence.

The counterexample answers the unrestricted question negatively. It does not weaken Corollary 7 of the source: additional metrizability or Frechet hypotheses still give the positive conclusion established there. Human review should focus on whether the phrase “a version of Theorem 5” is intended with the printed hypotheses only; under that natural reading, every hypothesis is met.

5 Bounded novelty check

On 26 August 2026, the four lightweight indexes of run `fa_banach_001` were searched for the arXiv id, title, and operator notation. Exact-phrase, title, and close-variant arXiv/web searches were also performed. Two related later arXiv papers on locally solid operator topologies and Lebesgue operators were checked at title/abstract level [2, 3]; neither was found to state or resolve this exact question. No later answer was found within these bounded searches. This is evidence of plausible novelty, not a certification.

References

- [1] O. Zabeti, *AM-spaces from a locally solid vector lattice point of view with applications*, arXiv:1912.00141; Bull. Iran. Math. Soc. 47 (2021), doi:10.1007/s41980-020-00458-7.
- [2] Y. Deng and M. de Jeu, *Convergence structures and locally solid topologies on vector lattices of operators*, arXiv:2008.05379.
- [3] S. Alpay, E. Emelyanov, and S. Gorokhova, *The σ -continuous, Lebesgue, KB, and Levi operators between vector lattices and topological vector spaces*, arXiv:2105.01810.